Subject: R. Helmbold Memo "Description of the Map Grid Used in the Ardennes Data Base (ACSDB)" Dated 22 Oct 91

The subject memo develops the basic equations required to convert geographic coordinates expressed in the ACSDB to latitude and longitude. The important points in the memo describing the conversion can be summarized as follows:

1. The ACSDB coordinates must first be converted to a Cartesian (x,y) system where x is km east and y is km north (of a specified origin point). This (x,y) system is denoted as "E (Easting) and N(Northing) offsets in km", or, equivalently as "Unigrid coordinates", in the memo. The equations to convert ACSDB locations to this (x,y) system are given on pgs 6-7 of the memo, and an example is presented there. The way the 1-letter prefixes fit together in a map is shown in pg 5 and Encl 1 of the memo. Encl 1 of the memo also gives the latitude-longitude of the origin in the (x,y) system.

2. After conversion to (x,y), (also denoted as Unigrid in the memo),the equations for converting to standard latitude-longitude are given in Encl 5 in matrix form. The equation form can be expressed as:

Longitude = .01389*x - .000188*y +4.9591

Latitude = .0000973* x + .00899*y +49.4805

where x is the Easting (X) coordinate and y is the Northing coordinate in the Unigrid (x,y) system.

The inverse relationships are also given in Encl 5.

The above is the heart of the conversion and can be programmed for computerized conversion.

The user should check any test calculations of the above against the worked-out examples given in Encl 4 of the memo.